

```
#include <graphics.h>
```

```
#include <conio.h>
```

```
#include <stdio.h>
```

```
#include <math.h>
```

```
void main()
```

```
{
```

```
    int gd = DETECT, gm;
```

```
    float x1, y1, x2, y2;
```

```
    float dx, dy, steps, xinc, yinc, x, y;
```

```
    int i;
```

```
    // Initialize graphics mode
```

```
    initgraph(&gd, &gm, "C:\\\\TURBOC3\\  
\\BGI");
```

```
    // Input coordinates
```

```
    printf("Enter x1, y1: ");
```

```
    scanf("%f %f", &x1, &y1);
```

```
    printf("Enter x2, y2: ");
```

```
scanf("%f %f", &x2, &y2);
```

```
// Calculate dx and dy
```

```
dx = x2 - x1;
```

```
dy = y2 - y1;
```

```
// Find the number of steps
```

```
if (fabs(dx) > fabs(dy))
```

```
    steps = fabs(dx);
```

```
else
```

```
    steps = fabs(dy);
```

```
// Calculate increment values
```

```
xinc = dx / steps;
```

```
yinc = dy / steps;
```

```
// Starting point
```

```
x = x1;
```

```
y = y1;
```

```
// Draw the line
```

```
for (i = 0; i <= steps; i++)  
{  
    putpixel(round(x), round(y), WHITE);  
    x += xinc;  
    y += yinc;  
}
```

```
getch();  
closegraph();  
}
```